2026 SIOP Machine Learning Competition

Eight years of pushing the I-O frontier through open competition

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Three assumptions about the future

1 **AI capability keeps jumping in orders of magnitude**

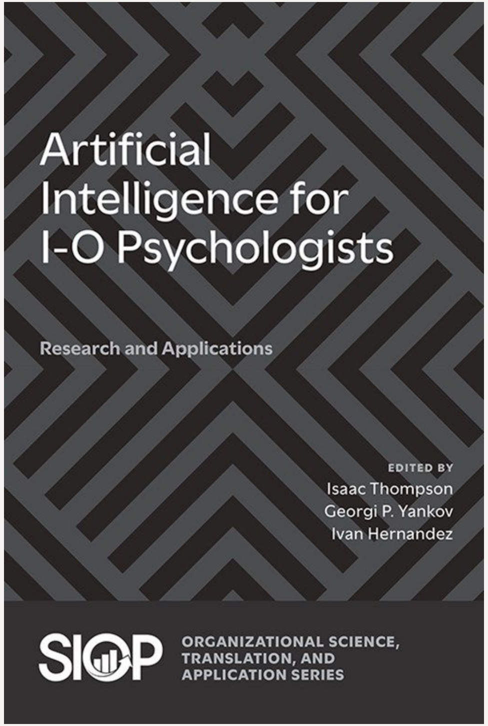
Each jump unlocks new task categories, not just better performance on old ones.

2 **Work and the economy get transformed alongside**

Repetitive cognitive work faces displacement on a compressed timescale.

3 **I-O psychology faces a double transformation**

Both methods (how to study work) and subject matter (what work is) shift at once.



Where does the real frontier live?

Three candidates; one earns the label

Academic research

Too slow

Publication bias and multi-year cycles. The frontier moves before the paper lands.

Business products

Too closed

Vendor incentives keep methods proprietary. Conflicts of interest filter what gets shared.

Open competition

The frontier

Empirical, fast, public. Solutions and data live past the competition and become benchmarks.

This competition has lived at that intersection for eight years.

What makes a competition frontier-spanning

Hard problem

Difficult enough that the answer is not obvious to the field

Open data

Gathered, cleaned, and shared with everyone on equal terms

Open code

Winning solutions released as open source after the competition

Open registration

Anyone can enter; no gatekeeping by affiliation or status

Objective scoring

A held-out test set, not a panel of judges, determines the winner

Durable artifacts

Data and code outlive the competition and become public benchmarks

How a machine learning competition works

Five stages

1. **Release**
   Training data and problem statement go public.
2. **Attempt**
   Community solves the problem empirically.
3. **Dev leaderboard**
   Public scoring on a development set tracks progress.
4. **Final**
   Held-out test set decides the winner. (private leaderboard)
5. **Benchmark**
   Data and winning solutions live on as public assets.

The benchmark stage is what compounds across years.

Eight years of pushing the I-O frontier

Each year showcased a novel problem and novel solutions that live past the competition

2018	Pure Empirical Methods to best Predict turnover	Eli Lilly & Co.
2019	Open ended text scoring (personality)	Modern Hire/ HireVue/ Amazon
2020 - 2021	New approaches to Optimize for Validity AND diversity	Walmart
2023	Open ended text scoring to automate assessment center scoring	DDI
2024	I-O benchmarks to Evaluate LLMs	HackerRank, Virginia Tech, DDI
2025	Game the System: AI to apply to jobs and beat selection assessment	Virgina Tech, Amazon
2026	AI to automate meta-analysis	**this year**

This Year

A realistic meta-analysis coding task: competitors must

- Identify the predictor – outcome effect size relevant to a research question
- Convert it to Pearson's *r* when needed
- Reverse-code relationships when constructs are reverse scored
- Return one aggregate effect size per study

Competitors design an end-to-end pipeline that accepts:

- PDF file name
- Research question description
- Predictor description
- Dependent-variable description

Competition Data

Research articles provided as PDFs

Each article is paired with a research question and construct descriptions:

- Research question description
- Predictor description
- Dependent-variable / criterion description

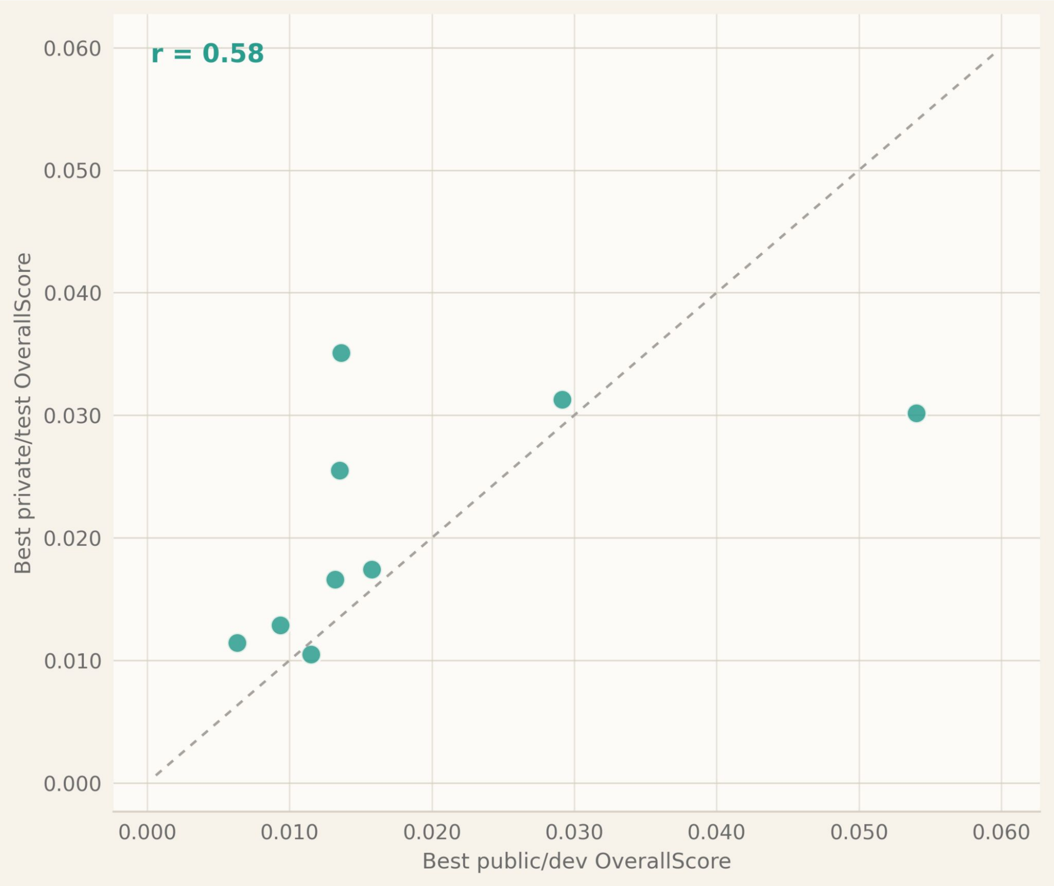
Some studies may contain multiple relevant effect sizes. The true study-level value is the average of the relevant observed correlations

Competitors submit a CSV with exactly two columns:

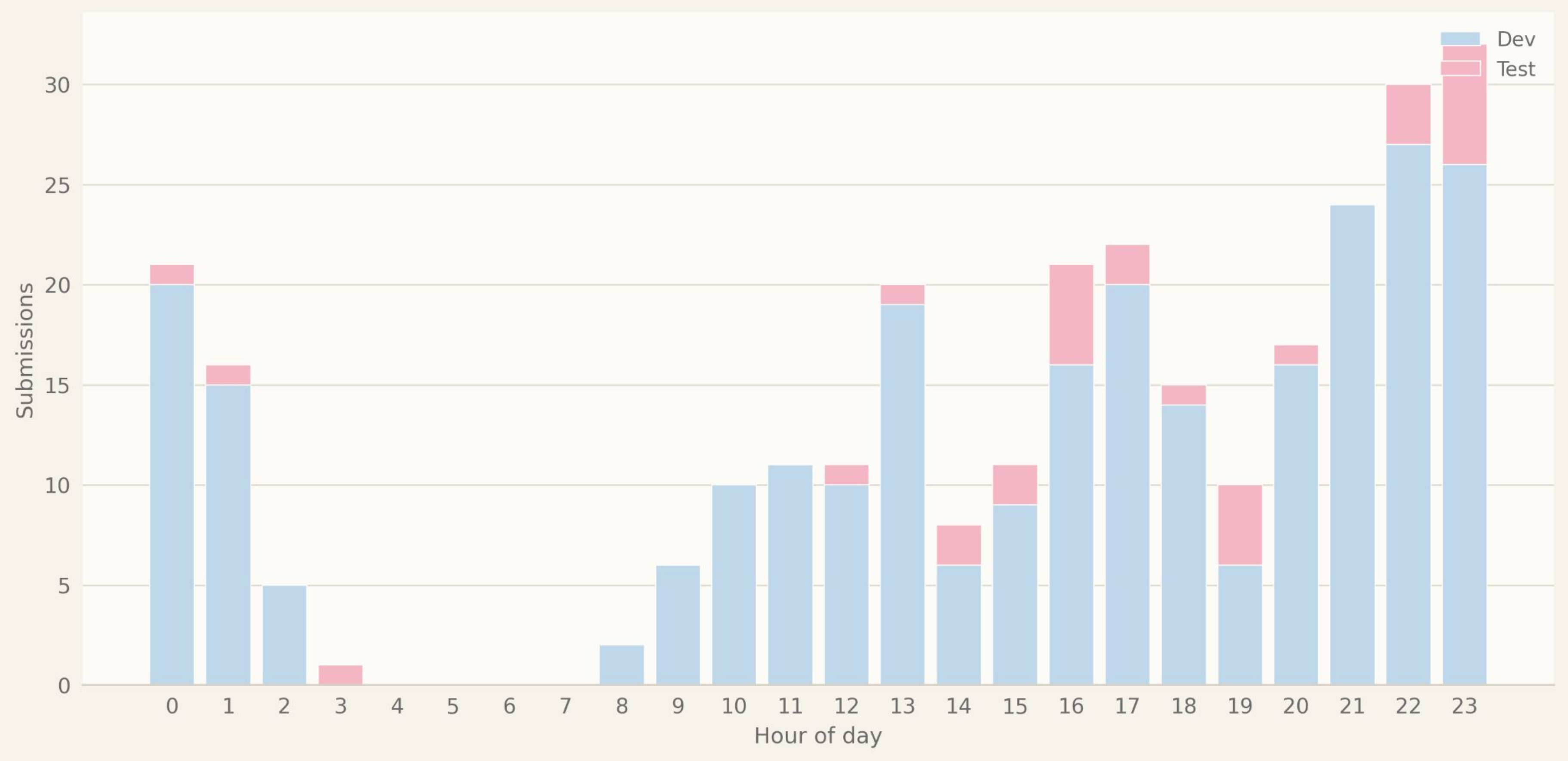
- StudyID
- Aggregate Effect Size

$$\text{Overall Score} = \frac{1}{n}\sum_{i=1}^{n}(\hat{r}_i - r_i)^2$$

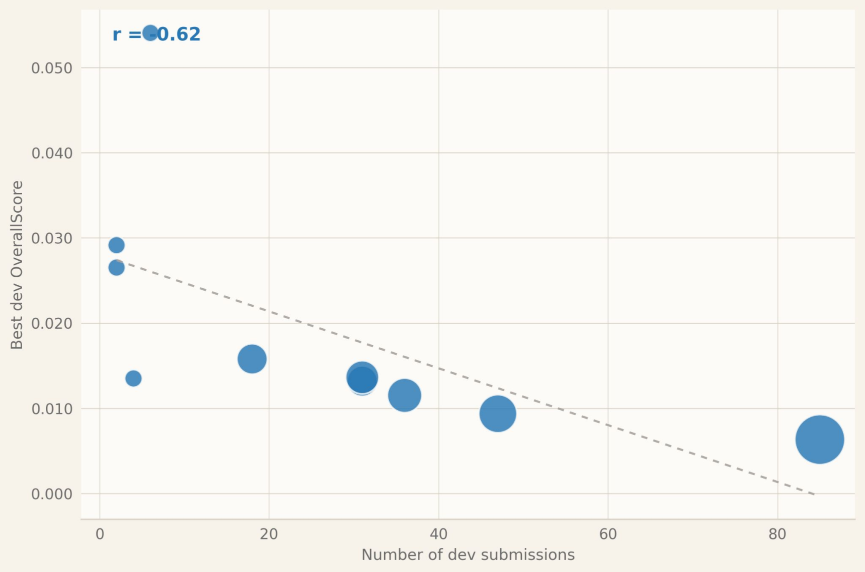
Correlation Between Dev & Test Performance



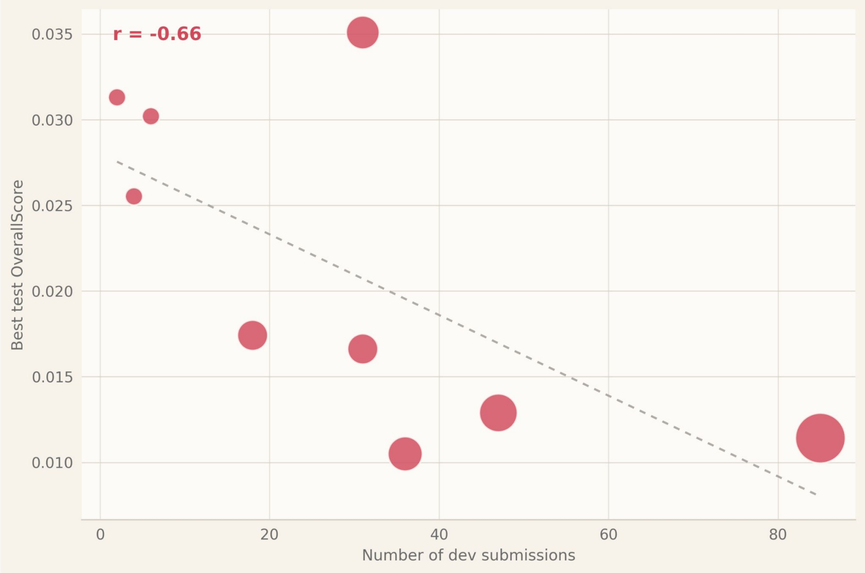
Submission Times



How # of Dev Submissions Relates to Model Error

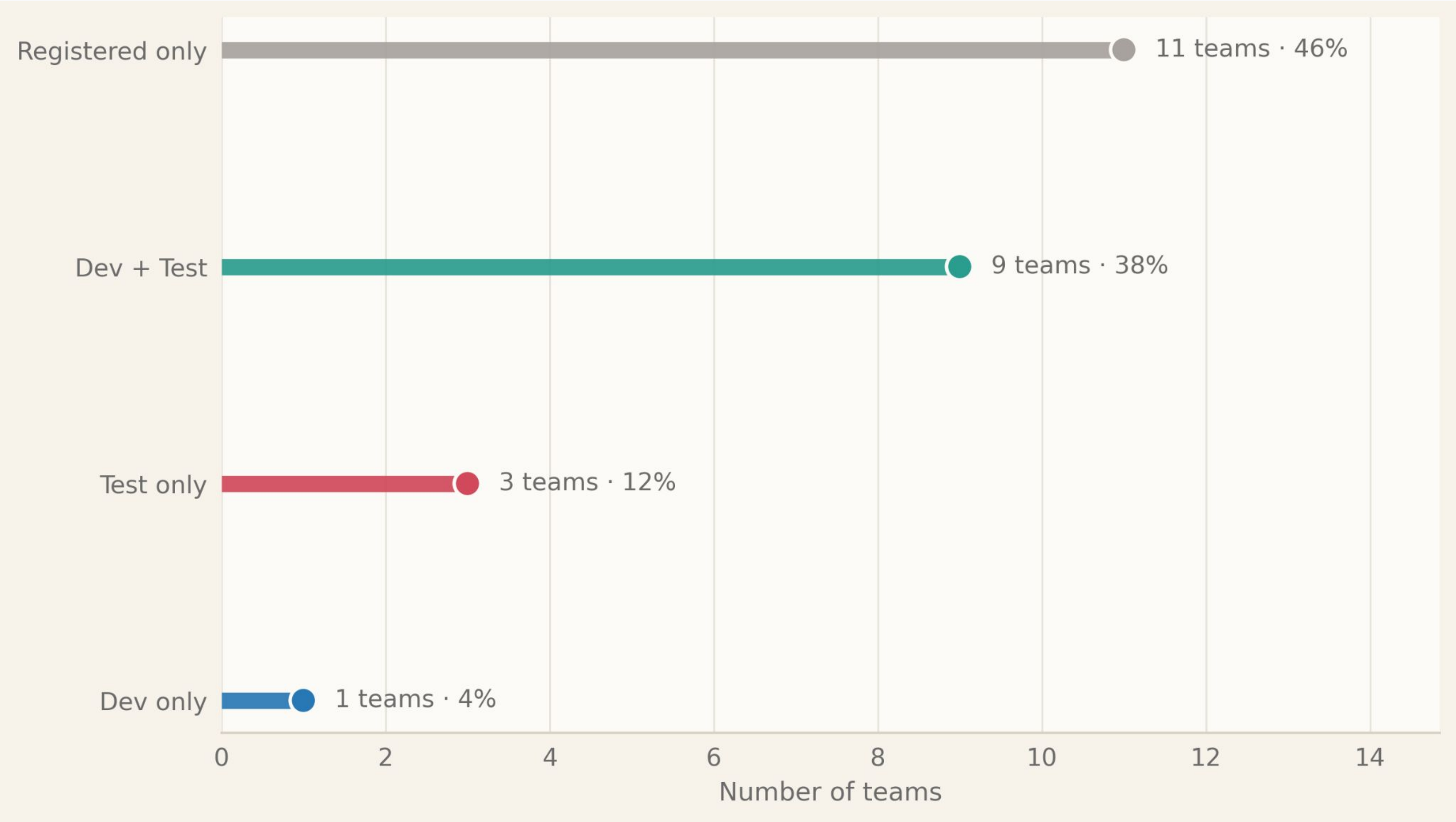


# of Dev Submissions and Development Error

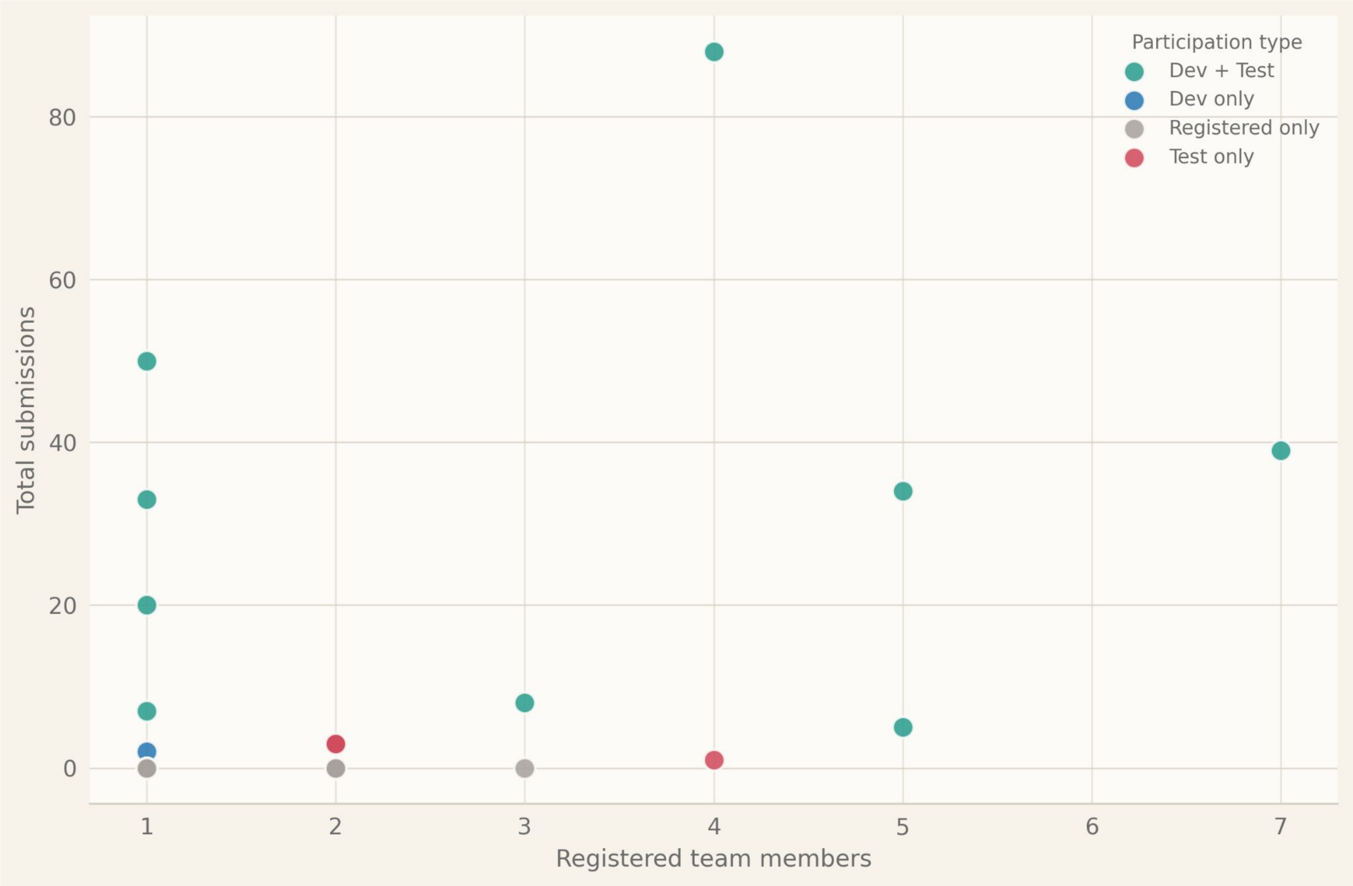


# of Dev Submissions and Test Error

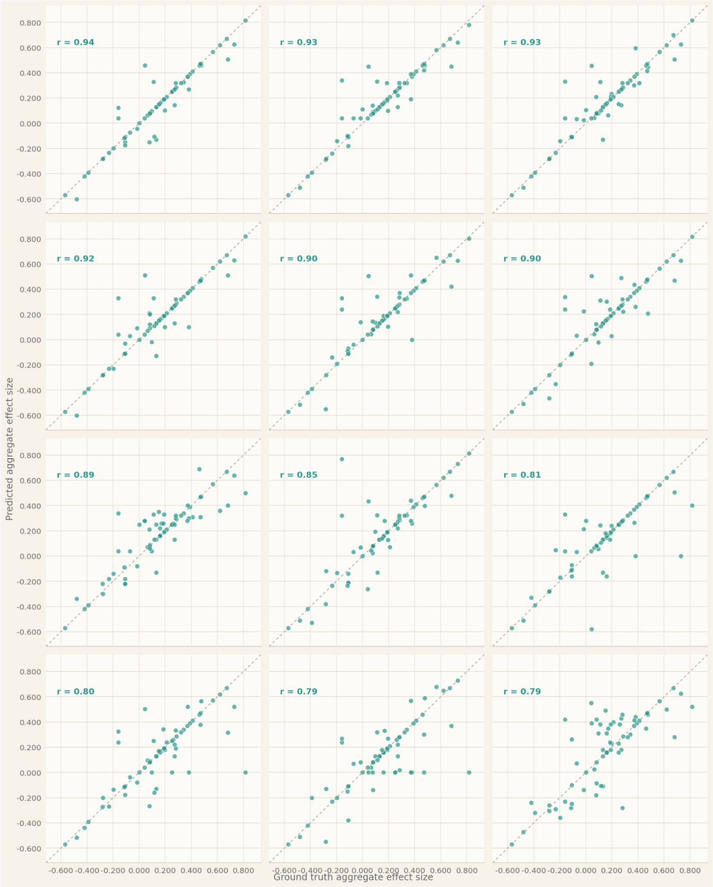
Type of Participation



Number of Submissions and Team Member Count



Correlation Between Model and Ground Truth



Correlation Between Model Prediction and Ground Truth Values

All Teams: r = 0.87

Top 4 Teams: r = 0.93

Correlation Between Model and Ground Truth

Mean Absolute Error Between Model Prediction and Ground Truth Values

All Teams: *r* = 0.07

Top 4 Teams: *r* = 0.05

Mean Absolute Error Between Model Prediction Across All Studies and Overall Effect Size

All Teams: *r* = 0.02

Top 4 Teams: *r* = 0.01

Top 12 Teams on Private/Test Leaderboard

Rank	Team	Final MSE
1	????	????
2	????	????
3	????	????
4	????	????
5	HubMeta	0.015928
6	ILMNinsights	0.016617
7	DSTST	0.017411
8	STATS Lab at CMC	0.025519
9	Neural Nomads	0.030193
10	Chen-Chillas	0.032653
11	One Hot Key	0.03508
12	Job Analysis	0.03623

The Finalists

The Top 4 Teams

(From 4th - 1st)

Fourth Place

MetaML

Final Score = 0.012888

Correlation with Ground Truth: 0.920927
Mean Absolute Error: 0.050972
Difference from Overall Effect: 0.010942

Team Members:

Pengda Wang

RICE

Solution Code

Third Place

Hungry Llama

ForsMarsh

Final Score = 0.011416

Correlation with Ground Truth: 0.929885
Mean Absolute Error: 0.050074
Difference from Overall Effect: 0.013898

Team Members:

Jennifer Gibson
Joe Luchman
Shane Halder
Ron Vega

Solution Code

Second Place

Putka’s Army

Final Score = 0.010497

Correlation with Ground Truth: 0.934952
Mean Absolute Error: 0.045669
Difference from Overall Effect: 0.013972

Team Members:

Ammar Ansari
Daniel Barstow
Anoop Javalagi
Jiayi Liu
Karla Castillo-Guerra
John Little
Robert Wellman
Lilang Chen

HUMAN RESOURCES RESEARCH ORGANIZATION

Solution Code

First Place

goforit

Final Score = 0.009233

Correlation with Ground Truth: 0.944140
Mean Absolute Error: 0.041927
Difference from Overall Effect: 0.006750

Team Members:

Nga Do

Michael Hazboun

UNIVERSITY OF MINNESOTA

Solution Code

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1	goforitL 🥇	0.009233
2	Putka's Army 🥈	0.010497
3	HungryLlama 🥉	0.011416
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